

RESEARCH ARTICLE

Robust group variable screening based on maximum Lq-likelihood estimation

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Variable screening plays an important role in ultra-high-dimensional data analysis. Most of the previous analyses have focused on individual predictor screening using marginal correlation or other rank-based techniques. When predictors can be naturally grouped, the structure information should be incorporated while applying variable screening. This study presents a group screening procedure that is based on maximum Lq-likelihood estimation, which is being increasingly used for robust estimation. The proposed method is robust against data contamination, including a heavy-tailed distribution of the response and a mixture of observations from different distributions. The sure screening property is rigorously established. Simulations demonstrate the competitive performance of the proposed method, especially in terms of its robustness against data contamination. Two real data analyses are presented to further illustrate its performance.

KEYWORDS

data contamination, dimensionality reduction, grouped variables, robustness

1 | INTRODUCTION

Data with ultrahigh dimensions, that is, the dimensions of predictors are much larger than the sample size, have been frequently collected in many fields, such as biomedical imaging, microarrays, and clinical medicine. With such data, it is difficult to directly apply variable selection methods based on penalized likelihood due to concerns of computational complexity and algorithmic instability. To address these challenges in linear regression, variable screening has been commonly used to reduce the dimensionality to a moderate scale. For example, when analyzing the lung adenocarcinoma (LUAD) data from the Cancer Genome Atlas (TCGA), rough screening methods have been conducted before using variable selection methods to identify the important genetic effects on the lung cancer.^{1,2} However, as studies have shown that genes can form groups based on the pathways they belong to, the most frequently used screening approaches, such as sure independence screening,³ are insufficient for genetic studies due to not incorporating the grouping structures of genes. Thus, a more focused effort is required to accommodate grouped predictors in variable screening.

Yang Li and Rong Li contributed equally to this work.

Actually, in many fields, predictors can be naturally grouped. This can be attributed to their natural classification or is based on certain similarity measures. An important example arises in genetic association studies, wherein cellular processes usually affect sets of genes acting in concert. It has been demonstrated that gene set enrichment analysis (GSEA) is a powerful analytical method that focuses on gene sets, and it is superior to single-gene analysis because it adds important effects on pathways.^{4,5} In addition, groups can also be generated through a common statistical analysis. For instance, non-parametric additive regression is often used to address the nonlinear relationship between the predictor and the response. In this method, each nonparametric component can be approximated by a linear combination of basis functions, which can be viewed as a group accordingly. For the sake of accuracy and interpretability, there is a need for screening methods that take the group structure into account.

The existing group screening methods can be regarded as an extension of individual variable screening using two approaches. One is to rank the cumulative contribution of the predictors in each group, wherein the individual contribution is measured by the marginal correlation between each predictor and the outcome. Most rank-based screening methods can be extended to group screening in this manner; details can be found in Qiu and Ahn.⁶ The other is to establish the utility of the variables in each group in relation to the response and to use the joint effect as a screening criterion. This is anticipated to be preferable to the first approach because it can better account for the collinearity or high correlations of the variables within a group; in contrast, regression with an individual variable introduces possibly much bigger bias especially for the marginally unimportant but jointly important variables in a group. Kang et al⁷ gives a simple example to illustrate this advantage of groupwise analysis. The second strategy has been considered by Qiu and Ahn⁶ and Niu et al,⁸ wherein they used the residual sum of squares and modified R^2 after adjusting for different group sizes based on least square estimation as the screening criteria, respectively.

However, it is well known that the least square estimation is not robust against outliers, especially for ultra-high-dimensional predictors. This is because the maximum spurious correlation between the predictors and the realized noise can be large in these cases.⁹ Therefore, robustness against data contamination, which commonly includes a heavy-tailed distribution of the response and a mixture of observations from different distributions, is critical in variable screening. For individual variable screening, many robust methods based on the robust marginal correlation have been proposed, such as Kendall τ correlation,¹⁰ distance correlation,¹¹ quantile correlation,¹² and model-free feature screening.^{13,14} However, these methods are not easy to extend for group screening using the second approach. As a result, this serves as the motivation for this study to develop a new group screening method that accounts for the group structure and data contamination. We note that the robustness of the screening depends, to a great extent, on the robustness of the estimation. We applied the maximum Lq-likelihood estimation (MLqE)¹⁵ to perform a groupwise regression owing to its appealing robustness against outliers.

In this study, we present a robust screening approach for ultra-high-dimensional data with grouped variables. It conducts rank screening by incorporating grouped information and a joint relationship between the group and the outcome, which can be applied to the case where the predictors are highly correlated. In addition, the sure screening property is established for the proposed screening method, and a numerical investigation provides strong evidence of the robustness of this method.

The rest of this article is organized as follows. We introduce the group screening method based on MLqE, hereinafter called Lq group screening, and present its theoretical property in Section 2. The numerical studies conducted to compare the performances of the proposed method and some alternative methods are described in Section 3. Section 4 presents the analysis of real data to confirm its usefulness, and the concluding remarks are provided in Section 5.

2 | METHODS

2.1 | Lq group screening

Let us begin with a review of MLqE. Suppose $\{X_1, \dots, X_n\}$ is a sample from a distribution with the probability density function $f(x; \theta)$, $\theta \in \Theta$. MLqE is defined as

$$\hat{\theta} = \arg \max_{\theta \in \Theta} \sum_{i=1}^n L_q(f(X_i; \theta)), \quad (1)$$

where $L_q(u) = (u^{1-q} - 1)/(1 - q)$ for $q > 0$ and $q \neq 1$. In particular, when $q \rightarrow 1$, $L_q(u) \rightarrow \log(u)$. The distortion parameter q determines the similarity between the L_q function and the log function. The corresponding estimating equations have

the following form:

$$\sum_{i=1}^n \nabla_{\theta} \log(f(X_i; \theta)) f(X_i; \theta)^{1-q} = 0. \quad (2)$$

The estimating equation of MLqE is a weighted version of that of the classical maximum likelihood estimation (MLE), where the weight is related to the probability density function, that is, $f(X_i; \theta)^{1-q}$. The weight matrix is denoted as $\mathbf{W} = \text{diag}(w_1, \dots, w_n)$, where $w_i = f(X_i; \theta)^{1-q}$. The contribution of each observation to the likelihood function is allocated by the weight matrix $\mathbf{W}$.

When $q < 1$, the observations that disagree with the model receive a low weight and those that agree with the model receive a high weight in MLqE. In this manner, the role of the observations is slightly modified by means of the distortion parameter q . The estimation can be regarded as the minimization of the difference between the distribution in a parametric family and the modified true distribution, which is achieved by diminishing the role of extreme observations. In this sense, the MLqE provides a robust protection to the assumed parametric assumption. The distortion parameter q adjusts the weighting. When $q = 1$, MLqE is equivalent to MLE as the weights are equal. The closer q is to 1, the smaller the difference between the observation weight with a large likelihood and that with a small likelihood, that is, the more sensitive the MLqE is to outliers. As for the selection of q , there is presently no general method. However, MLqE is generally less sensitive to data contamination than MLE (to different degrees) when q is smaller than 1.¹⁶

MLqE is a new class of parametric estimators that are motivated from the nonextensive q -entropy in information theory. It can achieve robustness through parametric likelihood estimation instead of nonparametric estimation. Unlike nonparametric methods that do not involve finite dimensional assumptions of the model, MLqE is a parametric method but partially weakens the inflexibility of parametric modeling by the adjustment of the distortion parameter q . Under the linear regression framework, MLqE keeps the asymptotic optimality of MLE (as $q \rightarrow 1$) while providing robust protection to the assumed parametric assumption with the use of the distortion parameter q . Additionally, in comparison to nonparametric methods that involve complex forms of expression, MLqE can reduce the computational complexity.

We now consider a linear regression model with p predictors. Suppose that the predictors can be divided into J nonoverlapping groups. The linear model can be written as follows:

$$Y = \sum_{j=1}^J \mathbf{X}_j \boldsymbol{\beta}_j + \epsilon, \quad (3)$$

where $Y = (Y_1, \dots, Y_n)^T$ is an $n \times 1$ response vector, $\mathbf{X}_j = (X_{j1}, \dots, X_{jp_j})$ is an $n \times p_j$ random design matrix of p_j predictors in the j th group, $\boldsymbol{\beta}_j = (\beta_{j1}, \dots, \beta_{jp_j})^T$ is the corresponding coefficient vector in the j th group, and $\epsilon = (\epsilon_1, \dots, \epsilon_n)^T$ is the i.i.d. random error with mean zero. The expression $\mathcal{M}_S$ denotes the true model, where $S = \{j : \boldsymbol{\beta}_j \neq \mathbf{0}, j = 1, \dots, J\}$. Due to the high dimension of the predictors, it is impossible to provide an estimator with the entire design matrix $\mathbf{X} = (\mathbf{X}_1, \dots, \mathbf{X}_J)$. Alternatively, we perform J low-dimensional marginal maximum Lq-likelihood estimates, according to the groups.

For the regression model with variables in the j th group $Y = \mathbf{X}_j \boldsymbol{\beta}_j + \eta_j$, denote the MLqE of the coefficient vector as $\hat{\boldsymbol{\beta}}_j, j = 1, \dots, J$. In particular, when the error term η_j has the distribution $N(0, \sigma^2)$,

$$f(Y_i | \mathbf{X}_{j,i}) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{1}{2\sigma^2} (Y_i - \mathbf{X}_{j,i}^T \boldsymbol{\beta}_j)^2 \right\},$$

where $\mathbf{X}_{j,i}^T$ is the i th row of the design matrix $\mathbf{X}_j, i = 1, \dots, n$. By solving the estimating equations (2), we obtain

$$\hat{\boldsymbol{\beta}}_j = (\mathbf{X}_j^T \mathbf{W} \mathbf{X}_j)^{-1} \mathbf{X}_j^T \mathbf{W} \mathbf{Y}, \quad (4)$$

$$\hat{\sigma}^2 = \frac{1}{\sum_{i=1}^n w_i} \sum_{i=1}^n w_i (Y_i - \mathbf{X}_{j,i}^T \hat{\boldsymbol{\beta}}_j)^2, \quad (5)$$

where $\mathbf{W} = \text{diag}(w_1, \dots, w_n)$, and $w_i = \left[\frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{1}{2\sigma^2} (Y_i - \mathbf{X}_{j,i}^T \hat{\boldsymbol{\beta}}_j)^2 \right\} \right]^{1-q}$. Since $\mathbf{W}$ depends on $\hat{\boldsymbol{\beta}}_j$, a simple iterative algorithm is available for solving (4) and (5). The details are presented in Algorithm 1. Let $\hat{\boldsymbol{\beta}}_j(0)$ and $\hat{\sigma}^2(0)$ be the maximum

Algorithm 1. MLqE of the regression model of the j th group**Input:** response Y and the random design matrix of the j th group $\mathbf{X}_j$ **Output:** $\hat{\beta}_j$ Initialization. $t = 0$ and $\|\hat{\beta}_j(t) - \hat{\beta}_j(t-1)\|_2 = \infty$.**for** $\|\hat{\beta}_j(t) - \hat{\beta}_j(t-1)\|_2 > \Delta$ **do** Update $t = t + 1$; Update $w_i(t), i = 1, \dots, n$. $w_i(t) = \left[\frac{1}{\sqrt{2\pi\hat{\sigma}^2(t-1)}} \exp \left\{ -\frac{1}{2\hat{\sigma}^2(t-1)} (Y_i - \mathbf{X}_{j,i}^T \hat{\beta}_j(t-1))^2 \right\} \right]^{1-q}$; Update $\hat{\beta}_j(t)$ with updated $\mathbf{W}$. $\hat{\beta}_j(t) = (\mathbf{X}_j^T \mathbf{W} \mathbf{X}_j)^{-1} \mathbf{X}_j^T \mathbf{W} \mathbf{Y}$; Update $\hat{\sigma}^2(t)$. $\hat{\sigma}^2(t) = \frac{1}{\sum_{i=1}^n w_i(t)} \sum_{i=1}^n w_i(t) \left(Y_i - \mathbf{X}_{j,i}^T \hat{\beta}_j(t) \right)^2$.**end for**

likelihood estimator as the initial value. In the algorithm, $\hat{\beta}_j(t)$ is the estimate of $\hat{\beta}_j$ for the t th iteration, and Δ is a tiny positive number, which is set to 10^{-6} in the simulation studies in Section 3.

In the framework of MLqE, the weight w_i in matrix $\mathbf{W}$ is proportional to the power transformation of the probability density, that is, $w_i = f(Y_i | \mathbf{X}_{j,i})^{1-q}$. Thus, the choice of w_i depends on the distribution assumption. In this study, the weights with normal distribution provide robust protection to the normal assumption in the sense that the MLqE will focus more on the data points around the center while weakening the data points on the tails with an appropriate $q < 1$, which in accordance with the mean regression of our interest. In practice, we recommend the use of normal distribution due to its satisfactory robustness and computational advantages in our numerical studies.

By obtaining the MLqE of the coefficients of regression using each group of variables, $\hat{\beta}_j, j = 1, \dots, J$, we can define a vector $\hat{\beta}^g = (\hat{\beta}_1^g, \dots, \hat{\beta}_J^g)$ to indicate the relative importance of the different groups, where

$$\hat{\beta}_j^g = p_j^{-1} \|\hat{\beta}_j\|_2 = p_j^{-1} \sqrt{(\hat{\beta}_{j1}^2 + \dots + \hat{\beta}_{jp_j}^2)}. \quad (6)$$

With these statistics, we can rank the joint effects of different groups by the L_2 norm of the estimated coefficients vectors. In fact, $\hat{\beta}_j^g = 0$ indicates that $\beta_{j1} = \dots = \beta_{jp_j} = 0$ in the j th group. For group screening, we adopt a simple strategy of ranking the components of $\hat{\beta}^g$ and selecting the largest d components. More precisely, let d be the number of groups that are retained after screening. We choose a submodel $\hat{\mathcal{M}}_d$ as

$$\hat{\mathcal{M}}_d = \left\{ \mathbf{X}_j : \hat{\beta}_j^g \text{ are among the largest } d \text{ of all } \hat{\beta}_j^g \text{s} \right\},$$

or for some $\gamma > 0$

$$\hat{\mathcal{M}}^\gamma = \left\{ \mathbf{X}_j : \hat{\beta}_j^g \geq \gamma \right\}.$$

2.2 | Theoretical property

We explore the theoretical performance of the proposed method, focusing on the sure screening property. Note that

$$Y = \sum_{j=1}^J \mathbf{X}_j \beta_j + \epsilon = \sum_{j=1}^J \sum_{k=1}^{p_j} X_{jk} \beta_{jk} + \epsilon,$$

which can be written as $Y = \mathbf{X}\beta + \epsilon$ by denoting $\mathbf{X} = (\mathbf{X}_1, \dots, \mathbf{X}_J)$ and $\beta = (\beta_1^T, \dots, \beta_J^T)^T$. Define

$$\mathbf{Z} = \mathbf{X}\Sigma^{-1/2},$$

where Σ is the $p \times p$ covariance matrix of the predictors. The n rows of the transformed design matrix $\mathbf{Z}$ are i.i.d. copies of a random vector with the covariance matrix I_p . We introduce the following assumptions:

- (A1) $E(\epsilon) = 0$, $\text{Var}(\epsilon) = \sigma^2$, and ϵ is independent of predictors $\mathbf{X}$.
 (A2) There are some $c_1 > 1$ and $C_1 > 0$, for any $n \times \tilde{p}$ submatrix $\tilde{\mathbf{Z}}$ of $\mathbf{Z}$, such that

$$P \left\{ \lambda_{\max}(\tilde{\mathbf{p}}^{-1} \tilde{\mathbf{Z}} \tilde{\mathbf{Z}}^T) > c_1 \text{ or } \lambda_{\min}(\tilde{\mathbf{p}}^{-1} \tilde{\mathbf{Z}} \tilde{\mathbf{Z}}^T) < c_1^{-1} \right\} \leq \exp(-C_1 n).$$

- (A3) $\text{Var}(Y) < \infty$, and there are some $\tau \geq 0$, $\kappa \geq 0$ and $c_2, c_4 > 0$, $0 < c_3 < 1$ such that

$$\min_{j \in S} p_j^{-1} \|\beta_j\|_1 \geq \frac{c_2}{n^\kappa}, \lambda_{\min}(\Sigma) \geq c_3, \lambda_{\max}(\Sigma) \leq c_4 n^\tau.$$

- (A4) There is some $M_1 > 1$ such that $\lambda_{\max}(\mathbf{W}) \leq M_1$ and $\lambda_{\min}(\mathbf{W}) \geq M_1^{-1}$.
 (A5) The number of true important groups $s \leq c_5 n^\nu$, $c_5 > 0$ for some $\nu \geq 0$ and the group size $p_j \leq c_6 n^l$ for $j = 1, \dots, J$ and $\iota \geq 0$, $c_6 > 0$, $\log(J) = O(n^\delta)$, $\delta \in (0, 1 - 2\kappa - 4\tau - \nu - \iota)$ with $2\kappa + 4\tau + \nu + \iota < 1$.

Theorem 1. Suppose that Assumptions (A1) to (A5) hold. If we choose γ_n such that

$$\frac{\gamma_n}{n^{1-\kappa-\tau-3/2\iota}} \rightarrow 0, \quad \frac{\gamma_n \sqrt{\log(n)}}{n^{1-\kappa-\tau-3/2\iota}} \rightarrow \infty,$$

then, for some constant C , we have

$$P \left(\max_{j \notin S} \hat{\beta}_j^g < \gamma_n < \min_{j \in S} \hat{\beta}_j^g \right) = 1 - O \left\{ \exp \left(-C \frac{n^{1-2\kappa-4\tau-\nu-\iota}}{\log(n)} \right) \right\}. \quad (7)$$

This implies that if we choose a submodel $\hat{\mathcal{M}}_d$ with $d > s$, we have

$$P(\mathcal{M}_S \subset \hat{\mathcal{M}}_d) = 1 - O \left\{ \exp \left(-C \frac{n^{1-2\kappa-4\tau-\nu-\iota}}{\log(n)} \right) \right\}. \quad (8)$$

Assumption (A2) is the property C defined in Fan and Lv,³ and it holds for a wide class of spherically symmetric distributions. Assumption (A4) is to limit the weight of the observations w_i , which can be achieved if the density function of the response variable $f(Y_i)$ is bounded and it has a bounded support. Although the bounded support cannot cover a wide distribution range, w_i will not be too small for most distributions. We can set a tiny positive number to limit w_i to prevent it from being too small in practical applications. According to this assumption, the tail behavior of the random error can be obtained, which would draw an analogy to the sub-Gaussian random variables. The details can be found in the supplementary material. The other assumptions are similar to those in Qiu and Ahn,⁶ especially for (A3). This assumption is for the grouped effect, that is, the average L_1 norm for the coefficient vectors of the important groups in (A3). The theorem states that under these conditions, the important and unimportant groups can be divided based on the threshold of the MLqE estimators in the regression model for each group. Thus, if we screen out at least s groups, the probability of including all the important groups is close to 1.

Remark: The sure screening property requires the assumptions on the average signal strength of the active groups, the correlation structure of the design matrix, the number of true important groups and the group size, and those conditions constrain each other in assumption (A5). Therefore, it may be necessary to consider the applicability of these conditions in specific complicated practical cases. More examples can be found in Section 3.

2.3 | Screening for nonparametric estimation

In addition to the cases of the naturally grouped observed variables, the dummy variables of a categorical variable or a set of spline bases for additive modeling also form a group structure. For nonparametric regression problems, Fan et al¹⁷ proposed nonparametric independence screening by applying a marginal nonparametric regression using a B-spline basis,

and ranking the importance according to the marginal magnitude. Using B-spline approximation, He et al¹⁸ proposed quantile-adaptive sure independence screening (QaSIS) method for individual screening by ranking the nonparametric marginal quantile utility. In this manner, the LqG can also be used for screening nonparametric components using the B-spline expansion. Specifically, we consider a nonparametric regression model

$$Y = \sum_{j=1}^p m_j(X_j) + \epsilon,$$

where $X_j = (X_{1j}, \dots, X_{nj})^T$, $Y = (Y_1, \dots, Y_n)^T$ is the response, and $\epsilon = (\epsilon_1, \dots, \epsilon_n)^T$ is the random error with mean zero. We assume that each $m_j(X_j)$ has a mean of zero and the response Y also has a mean of zero. Each nonparametric component $m_j(x)$ can be approximated by the spline function, that is,

$$m_j(x) \approx \sum_{k=1}^K \beta_{jk} \Psi_{jk}(x), \quad j = 1, \dots, p,$$

where $\{\Psi_{jk}, k = 1, \dots, K\}$ is a normalized B-spline basis. Then, the marginal nonparametric estimation problem is transformed to a parametric regression estimation problem with a group of variables $\{\Psi_{jk}(X_j), k = 1, \dots, K\}$. The function can be estimated by $\hat{m}_j(x) = \sum_{k=1}^K \hat{\beta}_{jk} \Psi_{jk}(x)$, where $\hat{\beta}_{jk}$ is the MLqE for β_{jk} based on the groupwise regression. Finally, the selected variables are a set

$$\hat{\mathcal{M}}_d = \{1 \leq j \leq p : \|\hat{m}_j\|^2 \text{ are among the largest } d \text{ of all } \|\hat{m}_j\|^2\},$$

where $\|\hat{m}_j\|^2 = n^{-1} \sum_{i=1}^n \hat{m}_j(X_{ij})^2$, or equivalently, a set for some γ

$$\hat{\mathcal{M}}^\gamma = \{1 \leq j \leq p : \|\hat{m}_j\|^2 \geq \gamma\}.$$

3 | NUMERICAL RESULTS

3.1 | Group variable screening

This study considers two cases with different sample sizes, numbers of groups, and predictor dimensions. We set $(n, J, p) = (100, 200, 1000)$ in Case I and $(n, J, p) = (200, 2000, 10\,000)$ in Case II, and there are five predictors in each group for both cases. Several scenarios are considered to establish the merit of the proposed method. The source code of the proposed method is provided at <https://github.com/lirong672/Lq-group-screening>.

- S1. $\mathbf{X}$ is generated from $N(\mathbf{0}, \Sigma)$, which is within group correlation ρ_1 and between group correlation ρ_2 . The error term ϵ follows a normal distribution $N(0, \sigma^2)$, where $\sigma = 1$. The first three groups have nonzero coefficients. Similar to Fan and Lv,³ the nonzero β_{jk} s are independent as $(-1)^U(a + |z|)$, $j = 1, \dots, 3$, $k = 1, \dots, 5$, where $a = 4 \log(n)/\sqrt{n}$, $U \sim \text{Bernoulli}(0.4)$, and $z \sim N(0, 1)$.
- S2. ϵ follows the t distribution $t(2)$. The other settings are the same as in S1.
- S3. ϵ follows the Cauchy distribution and the other settings are the same as in S1. The Cauchy distribution has heavy probability tails to produce outliers in the response in this scenario.
- S4. The samples are generated from different distributions with a contamination ratio of 0.1, that is, 10% of the response Y comes from another linear regression model, in which the true coefficient vector is

$$4 \times (0.5, -1, 1, 1, 2, 4.5, 0.1, -1, 0.5, 0.5, -3, -0.4, 3.2, 0.5, 2.6, 0, \dots, 0).$$

The other settings are the same as in S1.

- S5. Not all variables within a group are important. The settings are the same as in S3, except that a sparsity also exists within each group, three predictors have nonzero coefficients, and two have zero coefficients.
- S6. The generation of design matrix $\mathbf{X}$ and error term ϵ is the same as S3. The first four groups are active, with three of each group have nonzero coefficients $\beta_{jk} = (-1)^U(a + |z|)$, $j = 1, \dots, 4$, $k = 1, 2, 3$ and two of each group have

nonzero coefficients with $\beta_{jk} = (-1)^U(a + |z|)/5, j = 1, \dots, 4, k = 4, 5$, where $a = 4 \log(n)/\sqrt{n}$, $U \sim \text{Bernoulli}(0.4)$ and $z \sim N(0, 1)$.

- S7. The generation of design matrix $\mathbf{X}$ and error term ϵ is the same as S3. The first four groups are truly active and a sparsity also exists within the group, with only one predictor has a nonzero coefficient in each active group. The nonzero coefficients are set as $\beta_{j1} = (-1)^U(a + |z|), j = 1, \dots, 4$, where $a = 4 \log(n)/\sqrt{n}$, $U \sim \text{Bernoulli}(0.4)$ and $z \sim N(0, 1)$.

For the proposed approach, we take $q = 0.9$, denoted by LqG. The results for other choices of q are quite similar. Details can be found in the supplementary material. We also considered the following alternative methods:

- (1) Lq1: estimate each β_{jk} in regression with only one predictor by MLqE and cumulate the group effect using (6).
- (2) gRRCS: extend the robust rank-based screening using the Kendall τ correlation proposed by Li et al.¹⁰ to the group version, that is, cumulate the correlation of each individual variable within a group using (6).
- (3) gHOLP: according to the same rules, extend the high-dimensional ordinary least squares projection screening proposed by Wang and Leng.¹⁹
- (4) gSIS: the group SIS proposed by Niu et al.⁸
- (5) QaSIS: screening by marginal group quantile utility with the quantile $\alpha = 0.5$. The quantile-adaptive sure independence screening (QaSIS) method in He et al.¹⁸ is an applicable individual screening method, which screens variables by ranking the nonparametric marginal quantile utility using B-spline approximation. Here, we extend it to the group version by directly using $\mathbf{X}_j$ instead of $\pi(\mathbf{X}_j)$ in equation (3.1) of He et al.¹⁸
- (6) the individual screening methods: SIS,³ RRCS,¹⁰ HOLP¹⁹ and PartS⁷ with the given group structure. For a fair comparison, we also consider the proposed method's performance in individual screening. In a similar manner as PartS, the LqG, developed originally for group screening, is adjusted to individual screening through the corresponding component of MLqE $\hat{\beta}_j$ in (4), and referred to as LqG-I. We also modify QaSIS in the same way as LqG-I, replacing MLqE with quantile regression, and referred to as QaSIS-I.

Note that the difference between gHOLP herein and the method in Qiu and Ahn⁶ is that the L_2 norm is used instead of the L_1 norm. As illustrated in Section 1, the first three alternatives are extended using the first approach, whereas LqG, gSIS and QaSIS are extended using the second approach. We expect that these three methods can perform well in the strong correlation cases. In addition, the individual screening methods are compared with the group screening methods in terms of screening the individual predictors in settings S6 and S7.

We use two criteria to evaluate the performance of these methods. The first is the minimum number of groups in a ranking that is required to include all of the truly active groups, which is referred to as MS. The second criterion is the proportion of the significant groups surviving after screening, which is referred to as PC. A screening procedure is valuable if MS is close to the true number of active groups and PC is close to 1. To ensure a fair comparison, we select the same size of $\lceil n/\log(n) \rceil$ groups for each method in settings S1-S5 and selected the same size of $5\lceil n/\log(n) \rceil$ individual predictors for each method in settings S6 and S7, which simply combines all predictors in the selected groups. The results are shown in Tables 1, 2, and 3. In each cell, the mean values (standard error in parentheses) are based on 100 replications. We set 10^{-4} to control the minimum of w_i for the bounded property in (A4). From the numerical results, the following conclusions were obtained:

- (1) In S1 with a normal distribution, all of the screening methods perform well when $\rho_1 = \rho_2 = 0$, and the proportion PC is close to 1 and MS is close to the true number of groups with nonzero coefficients.
- (2) We evaluated the performance under the nonnormally distributed error term in S2 and S3. When the predictors are independent, all of the methods have a large PC and MS that is close to the true number of active groups in S2. However, only the MLqE based methods and gRRCS have a PC close to 1 under the Cauchy distribution. In S4, in which the data contamination comes from the heterogeneity with 10% of the observations from another model, Lq1 and gRRCS have a better performance than gHOLP and gSIS in terms of the PC and MS. QaSIS and LqG have competitive performance, which suggests that these two methods are more robust against heterogeneity.
- (3) When the predictors are correlated, the second extension approach (LqG, gSIS, and QaSIS) shows a better performance than the first approach (Lq1 and gRRCS). In addition, gHOLP works well for the correlated predictors because it is proposed to relax the correlation restriction.

TABLE 1 Simulation results for Case I where $(n, J, p) = (100, 200, 1000)$ under S1-S5

	(ρ_1, ρ_2)	LqG		Lq1		gRRCS		gHOLP		gSIS		QaSIS	
		PC	MS	PC	MS	PC	MS	PC	MS	PC	MS	PC	MS
S1	(0, 0)	1.000	3.11	0.997	3.43	0.997	4.13	1.000	3.22	1.000	3.11	0.987	5.30
		(0.000)	(0.49)	(0.033)	(2.53)	(0.033)	(6.62)	(0.000)	(1.26)	(0.000)	(0.56)	(0.065)	(6.01)
	(0.4, 0.4)	1.000	3.42	0.863	27.90	0.853	30.32	1.000	3.24	0.997	3.73	0.980	8.24
		(0.000)	(1.12)	(0.217)	(40.16)	(0.223)	(41.20)	(0.000)	(1.44)	(0.033)	(3.39)	(0.079)	(14.06)
	(0.6, 0.2)	0.960	10.48	0.730	43.62	0.720	48.12	0.997	5.31	0.967	9.05	0.913	20.86
S2	(0, 0)	1.000	3.14	1.000	3.40	0.993	4.16	1.000	3.29	1.000	3.19	0.993	4.95
		(0.000)	(0.40)	(0.000)	(1.48)	(0.047)	(5.72)	(0.000)	(1.10)	(0.000)	(0.48)	(0.047)	(4.57)
	(0.4, 0.4)	1.000	3.71	0.850	30.00	0.843	32.75	1.000	3.47	1.000	3.92	0.973	8.35
		(0.000)	(1.28)	(0.218)	(39.88)	(0.218)	(41.63)	(0.000)	(1.87)	(0.000)	(1.72)	(0.090)	(11.25)
	(0.6, 0.2)	0.967	11.67	0.743	43.67	0.713	48.80	0.980	7.93	0.957	12.87	0.890	23.01
S3	(0, 0)	0.987	5.63	0.987	5.55	0.990	6.36	0.660	58.03	0.680	59.42	0.807	36.93
		(0.081)	(11.74)	(0.081)	(11.48)	(0.057)	(17.65)	(0.397)	(67.89)	(0.386)	(69.13)	(0.237)	(42.22)
	(0.4, 0.4)	0.980	6.85	0.813	31.57	0.803	36.18	0.600	68.14	0.620	74.75	0.880	23.06
		(0.114)	(13.46)	(0.264)	(42.11)	(0.241)	(42.13)	(0.408)	(69.19)	(0.392)	(70.76)	(0.238)	(39.82)
	(0.6, 0.2)	0.917	18.78	0.693	51.19	0.697	51.60	0.593	69.56	0.570	79.62	0.807	36.93
S4	(0, 0)	0.997	4.77	0.983	6.25	0.987	6.11	0.940	13.81	0.947	13.89	0.970	8.00
		(0.033)	(5.27)	(0.073)	(9.45)	(0.065)	(9.51)	(0.128)	(19.46)	(0.122)	(22.61)	(0.095)	(9.13)
	(0.4, 0.4)	0.983	8.38	0.900	23.93	0.893	24.02	0.770	44.10	0.767	51.66	0.957	10.58
		(0.073)	(14.85)	(0.197)	(38.23)	(0.194)	(37.54)	(0.229)	(43.82)	(0.243)	(50.82)	(0.122)	(14.64)
	(0.6, 0.2)	0.843	27.05	0.877	23.81	0.873	24.49	0.860	23.17	0.833	32.79	0.883	24.33
S5	(0, 0)	0.987	6.24	0.990	6.35	1.000	4.13	0.630	64.90	0.643	59.62	0.780	38.88
		(0.065)	(14.44)	(0.057)	(17.35)	(0.000)	(2.51)	(0.397)	(67.89)	(0.392)	(64.15)	(0.303)	(50.62)
	(0.4, 0.4)	0.947	13.28	0.607	106.36	0.627	96.28	0.577	75.50	0.543	77.14	0.830	34.98
		(0.147)	(20.35)	(0.179)	(56.76)	(0.196)	(57.96)	(0.400)	(69.71)	(0.379)	(62.47)	(0.285)	(51.15)
	(0.6, 0.2)	0.823	33.03	0.630	143.22	0.637	142.93	0.573	78.61	0.533	88.15	0.780	38.88

Note: The mean values (standard error in parentheses) are based on 100 replications.

- (4) Although gHOLP and gSIS can deal with the correlated predictors, they do not perform well in the Cauchy error distribution and the heterogeneity cases. Lq1 and gRRCS are robust, but they are negatively affected by the correlation between the variables. QaSIS is developed for identifying the important predictors for quantile of response, which seems appealing to accommodate heterogeneity in S4. Therefore, it appears that only LqG can achieve a desirable performance under the data contamination and the correlated predictors.
- (5) Screening becomes more difficult when there is sparsity in the groups. All methods perform worse in S5 than in S3. Nonetheless, the performance of LqG is better than that of the alternatives.
- (6) When $p = 10\,000$, LqG is significantly better than the other methods in terms of MS from the mean and deviation perspectives. It is demonstrated that the proposed method is more stable for the contaminated data than the alternatives when p is very large.

TABLE 2 Simulation results for Case II where $(n, J, p) = (200, 2000, 10000)$ under S1 to S5

	(ρ_1, ρ_2)	LqG		Lq1		gRRCS		gHOLP		gSIS		QaSIS	
		PC	MS	PC	MS	PC	MS	PC	MS	PC	MS	PC	MS
S1	(0, 0)	1.000	3.01	1.000	3.06	1.000	3.05	1.000	3.04	1.000	3.01	1.000	3.18
		(0.000)	(0.10)	(0.000)	(0.42)	(0.000)	(0.36)	(0.000)	(0.40)	(0.000)	(0.10)	(0.000)	(1.60)
	(0.4, 0.4)	1.000	3.06	0.897	72.22	0.877	98.50	1.000	3.04	1.000	3.28	1.000	3.39
		(0.000)	(0.60)	(0.168)	(173.00)	(0.187)	(231.34)	(0.000)	(0.40)	(0.000)	(2.79)	(0.000)	(2.51)
	(0.6, 0.2)	0.997	9.30	0.753	179.43	0.733	183.88	0.997	6.18	0.997	7.94	0.960	18.62
S2	(0, 0)	1.000	3.00	1.000	3.00	1.000	3.00	1.000	3.01	1.000	3.00	1.000	3.09
		(0.000)	(0.00)	(0.000)	(0.00)	(0.000)	(0.00)	(0.000)	(0.10)	(0.000)	(0.00)	(0.000)	(0.80)
	(0.4, 0.4)	1.000	3.00	0.900	94.09	0.883	106.26	1.000	3.03	1.000	3.16	0.993	6.29
		(0.000)	(0.00)	(0.180)	(232.29)	(0.191)	(240.33)	(0.000)	(0.30)	(0.000)	(1.03)	(0.047)	(19.45)
	(0.6, 0.2)	1.000	4.87	0.707	251.75	0.693	253.62	0.973	12.97	0.990	7.34	0.953	26.65
S3	(0, 0)	1.000	3.07	1.000	3.28	1.000	3.05	0.640	438.04	0.643	450.88	0.953	58.59
		(0.000)	(0.43)	(0.000)	(1.44)	(0.000)	(0.33)	(0.410)	(603.28)	(0.403)	(621.00)	(0.189)	(269.87)
	(0.4, 0.4)	0.997	4.41	0.813	129.55	0.803	126.28	0.533	556.86	0.533	596.15	0.927	112.32
		(0.033)	(10.53)	(0.237)	(227.81)	(0.231)	(194.57)	(0.424)	(658.29)	(0.424)	(665.41)	(0.224)	(376.46)
	(0.6, 0.2)	0.937	20.45	0.670	320.28	0.650	333.49	0.470	613.18	0.493	631.25	0.867	115.42
S4	(0, 0)	1.000	3.02	1.000	3.29	1.000	3.25	0.930	34.71	0.953	30.42	1.000	3.26
		(0.000)	(0.14)	(0.000)	(1.26)	(0.000)	(1.31)	(0.144)	(103.86)	(0.125)	(93.79)	(0.000)	(1.14)
	(0.4, 0.4)	0.997	3.84	0.973	18.71	0.983	19.88	0.740	243.33	0.683	403.10	0.993	5.97
		(0.033)	(5.59)	(0.102)	(74.88)	(0.073)	(104.31)	(0.248)	(375.11)	(0.247)	(537.28)	(0.047)	(19.62)
	(0.6, 0.2)	0.960	15.61	0.917	62.58	0.907	59.33	0.877	54.41	0.853	158.63	0.970	15.89
S5	(0, 0)	0.987	6.26	0.987	5.99	1.000	3.04	0.517	565.74	0.517	557.27	0.960	42.67
		(0.081)	(19.00)	(0.081)	(17.53)	(0.000)	(0.24)	(0.436)	(658.18)	(0.430)	(649.27)	(0.165)	(236.88)
	(0.4, 0.4)	0.977	24.71	0.527	1001.05	0.553	861.74	0.400	676.02	0.357	775.81	0.897	95.56
		(0.108)	(105.93)	(0.178)	(583.56)	(0.177)	(579.06)	(0.403)	(668.88)	(0.369)	(682.65)	(0.225)	(287.70)
	(0.6, 0.2)	0.887	68.07	0.597	1587.05	0.590	1590.79	0.360	723.70	0.387	872.17	0.927	77.47

Note: The mean values (standard error in parentheses) are based on 100 replications.

- (7) Generally speaking, group screening methods perform better than individual screening with larger true positive rate (PC) and smaller false positive rate (MS) in S6. This is reasonable since the weak significant predictors in the group are not easy to be screened out alone by individual screening. However, individual screening methods perform better than group screening with smaller MS and larger PC in S7, since the sparsity in the groups makes the group average signal too small to be screened out by group screening. Compared with the alternatives, the proposed LqG and LqG-I perform best in S6 and S7, respectively.

Besides, we compare the computation time for different screening methods under S3 with Cauchy error term. When the sample size is fixed, the computation time (in seconds) varying with the number of groups (J) is shown in Figure 1. As

TABLE 3 Simulation results under S6 and S7

		Case I: $(n, J, p) = (100, 200, 1000)$						Case II: $(n, J, p) = (200, 2000, 10\ 000)$					
		S6			S7			S6			S7		
(ρ_1, ρ_2)		(0, 0)	(0.4, 0.4)	(0.6, 0.2)	(0, 0)	(0.4, 0.4)	(0.6, 0.2)	(0, 0)	(0.4, 0.4)	(0.6, 0.2)	(0, 0)	(0.4, 0.4)	(0.6, 0.2)
LqG	PC	0.955 (0.129)	0.858 (0.188)	0.638 (0.243)	0.805 (0.199)	0.780 (0.219)	0.530 (0.210)	0.973 (0.136)	0.923 (0.186)	0.705 (0.219)	0.880 (0.192)	0.740 (0.255)	0.515 (0.226)
	MS	76.50 (136.83)	177.60 (194.56)	307.15 (225.44)	195.49 (174.37)	242.07 (212.33)	458.24 (238.73)	159.70 (798.03)	455.20 (1457.04)	1391.80 (1966.30)	543.65 (1332.14)	1190.52 (1914.47)	2258.74 (2017.91)
LqG-I	PC	0.504 (0.094)	0.323 (0.069)	0.330 (0.085)	0.945 (0.135)	0.780 (0.159)	0.818 (0.215)	0.495 (0.103)	0.266 (0.069)	0.241 (0.084)	0.950 (0.158)	0.730 (0.199)	0.755 (0.229)
	MS	882.01 (96.06)	944.05 (47.41)	919.31 (80.84)	68.00 (117.29)	314.56 (253.11)	228.01 (244.80)	8713.42 (1194.09)	9702.75 (285.68)	9069.45 (985.91)	275.89 (1069.00)	2686.06 (2637.96)	1008.47 (1628.45)
gRRCS	PC	0.965 (0.094)	0.575 (0.185)	0.618 (0.130)	0.840 (0.168)	0.550 (0.184)	0.665 (0.155)	0.995 (0.035)	0.510 (0.166)	0.648 (0.146)	0.893 (0.155)	0.458 (0.177)	0.678 (0.134)
	MS	72.85 (114.94)	696.55 (243.86)	884.30 (164.89)	184.85 (172.03)	712.35 (261.73)	740.93 (259.88)	36.95 (48.46)	7774.30 (2124.24)	9376.10 (1046.03)	246.54 (397.93)	8038.17 (2254.23)	8492.95 (2070.68)
RRCS	PC	0.522 (0.086)	0.362 (0.071)	0.517 (0.083)	0.950 (0.117)	0.725 (0.075)	0.765 (0.131)	0.496 (0.070)	0.308 (0.066)	0.502 (0.078)	0.980 (0.068)	0.720 (0.081)	0.753 (0.083)
	MS	890.00 (85.01)	994.55 (11.64)	961.12 (46.63)	60.42 (91.78)	971.31 (67.80)	465.91 (314.15)	8778.14 (1021.79)	9988.19 (22.94)	9857.16 (222.22)	74.09 (233.81)	9984.48 (34.67)	5392.61 (3432.51)
gHOLP	PC	0.635 (0.396)	0.578 (0.335)	0.590 (0.325)	0.445 (0.340)	0.425 (0.321)	0.518 (0.349)	0.443 (0.371)	0.358 (0.338)	0.358 (0.315)	0.280 (0.288)	0.200 (0.237)	0.285 (0.289)
	MS	328.55 (323.14)	408.45 (335.53)	424.45 (314.04)	509.34 (312.28)	558.01 (293.85)	498.26 (318.83)	3733.70 (3257.76)	4584.60 (3218.19)	4360.00 (3276.03)	5156.29 (3214.90)	5904.81 (2870.40)	5147.44 (3218.13)
HOLP	PC	0.347 (0.187)	0.311 (0.153)	0.417 (0.215)	0.578 (0.357)	0.588 (0.347)	0.613 (0.347)	0.186 (0.150)	0.137 (0.118)	0.264 (0.211)	0.410 (0.398)	0.308 (0.341)	0.375 (0.361)
	MS	923.58 (75.90)	925.20 (64.31)	886.22 (100.17)	429.11 (351.76)	462.42 (346.98)	417.87 (339.22)	9029.35 (922.61)	9248.65 (717.15)	8686.60 (1475.34)	4121.73 (3510.82)	4542.69 (3197.40)	4452.18 (3367.91)
gSIS	PC	0.628 (0.391)	0.493 (0.356)	0.398 (0.283)	0.423 (0.337)	0.430 (0.298)	0.300 (0.245)	0.480 (0.395)	0.333 (0.330)	0.208 (0.258)	0.273 (0.281)	0.193 (0.237)	0.120 (0.182)
	MS	350.30 (334.38)	455.50 (311.83)	553.40 (284.78)	531.63 (311.67)	569.61 (283.69)	626.41 (236.01)	3763.40 (3282.13)	4843.85 (3193.72)	5694.30 (2898.20)	5186.62 (3278.91)	5872.32 (2843.57)	6475.12 (2416.57)
SIS	PC	0.345 (0.174)	0.248 (0.102)	0.387 (0.174)	0.593 (0.353)	0.518 (0.253)	0.550 (0.304)	0.187 (0.152)	0.125 (0.098)	0.245 (0.170)	0.413 (0.401)	0.268 (0.272)	0.318 (0.295)
	MS	916.04 (70.50)	972.81 (45.80)	945.34 (60.56)	427.23 (345.83)	864.55 (169.82)	576.00 (310.31)	9009.57 (978.11)	9762.41 (355.29)	9442.35 (739.58)	4090.61 (3523.48)	8575.67 (1712.60)	6045.42 (2839.34)
PartS	PC	0.334 (0.175)	0.231 (0.110)	0.230 (0.111)	0.580 (0.379)	0.515 (0.291)	0.493 (0.340)	0.184 (0.154)	0.094 (0.079)	0.085 (0.081)	0.415 (0.402)	0.290 (0.287)	0.245 (0.281)
	MS	921.84 (74.33)	947.90 (51.70)	927.11 (75.12)	428.14 (339.12)	555.30 (292.66)	486.34 (315.37)	9104.52 (923.05)	9471.66 (424.49)	9363.06 (666.13)	4058.10 (3480.47)	5432.96 (3066.60)	5151.70 (3072.40)
QaSIS	PC	0.853 (0.227)	0.585 (0.232)	0.638 (0.204)	0.710 (0.208)	0.548 (0.196)	0.665 (0.283)	0.935 (0.186)	0.580 (0.220)	0.663 (0.185)	0.800 (0.245)	0.460 (0.214)	0.603 (0.212)
	MS	197.90 (259.33)	492.45 (264.23)	513.30 (293.48)	312.34 (242.52)	506.38 (272.29)	345.61 (302.71)	377.21 (1367.09)	4098.55 (2950.53)	4702.30 (3254.07)	1039.22 (1687.37)	4091.09 (2893.10)	4355.16 (3125.08)
QaSIS-I	PC	0.408 (0.114)	0.274 (0.081)	0.255 (0.097)	0.875 (0.211)	0.760 (0.183)	0.610 (0.246)	0.398 (0.237)	0.215 (0.079)	0.186 (0.086)	0.898 (0.206)	0.690 (0.218)	0.675 (0.244)
	MS	904.88 (98.63)	953.30 (47.03)	922.51 (74.18)	186.05 (256.21)	336.34 (254.96)	496.29 (297.18)	5076.28 (2732.34)	9595.38 (378.21)	9046.53 (834.57)	528.85 (1495.42)	2458.92 (2649.82)	1687.52 (2182.11)

Note: The mean values (standard error in parentheses) are based on 100 replications.

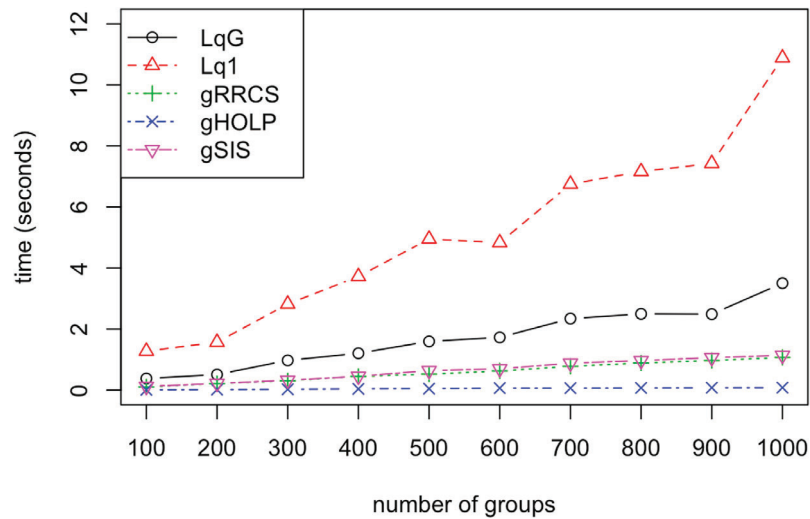


FIGURE 1 The computation time (in seconds) of each method vs the number of groups [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/sim.9212)]

we can see, gHOLP has the shortest computation time and Lq1 is the most time-consuming method. Although the MLqE based screening method takes slightly more time than others due to the iteration algorithm of MLqE, the computation time increases linearly as the number of groups increases. The proposed method only takes about five iterations and a few seconds even in ultra-high-dimensional cases. Overall, the computation cost seems worthwhile for LqG.

3.2 | Screening for nonparametric estimation

This section assesses the screening performance for the nonparametric model with simulated data. The data are generated from the following nonparametric model:

$$Y = 3g_1(X_1) + 4g_2(X_2) + 6g_3(X_3) + \epsilon,$$

where

$$g_1(x) = (2x - 1)^2, \quad g_2(x) = \frac{\sin(2\pi x)}{2 - \sin(2\pi x)},$$

$$g_3(x) = 0.1 \sin(2\pi x) + 0.2 \cos(2\pi x) + 0.3 \sin(2\pi x)^2 + 0.4 \cos(2\pi x)^3 + 0.5 \sin(2\pi x)^3.$$

The covariates are simulated according to $X_j = \frac{W_j + tU}{1+t}$, $j = 1, \dots, p$, where $W_1, \dots, W_p$, and U are i.i.d. from $\text{Unif}(0, 1)$. When $t = 0$, the covariates are all independent, and when $t = 1$, the pairwise correlation of the covariates is 0.5. Here, we consider different distributions of ϵ to evaluate the screening accuracy of each method for a variety of situations. We consider the normal distribution, the $\chi^2(2)$ distribution, and the Gaussian mixture distribution with the density function $f_\epsilon(x) = 0.8\phi(x; 0, 2) + 0.2\phi(x; 0, 36)$, where $\phi(x; \mu, \sigma^2)$ is the density function of $N(\mu, \sigma^2)$. For each method, we consider $K = \lceil n^{1/5} \rceil = 3$. We set two cases: $(n, p) = (100, 200)$ as Case I, and $(n, p) = (200, 2000)$ as Case II. In addition, the PC and MS criteria are used as the measurements.

The results are given in Table 4. Here, we consider the methods LqG, gRRCS, gHOLP, gSIS as well as QaSIS with quantile $\alpha = 0.5$. The Lq1 is removed because it is unreasonable for the nonparametric estimation to estimate each coefficient of the basis function separately. Note that gSIS is equal to the nonparametric screening indicated in Fan et al.¹⁷ It is observed that QaSIS has comparable PC to gSIS, and performs better than gSIS in terms of MS when ϵ follows the Gaussian mixture distribution, which is expected because the quantile regression is more robust against the heterogeneous data than the least squares regression. QaSIS also performs better than gRRCS when the predictors are correlated ($t = 1$). However, it performs worse than the proposed method LqG. The results indicate that LqG has the best performance and

TABLE 4 Simulation results in the nonparametric regression case ($p = J$)

ϵ	t	LqG		gRRCS		gHOLP		gSIS		QaSIS	
		PC	MS	PC	MS	PC	MS	PC	MS	PC	MS
Case I: $(n, p) = (100, 200)$											
Normal	$t = 0$	1.000 (0.000)	3.46 (2.03)	1.000 (0.000)	3.99 (2.93)	0.997 (0.033)	4.42 (4.27)	1.000 (0.000)	3.39 (1.63)	0.980 (0.079)	7.09 (9.95)
	$t = 1$	0.990 (0.057)	7.55 (15.41)	0.870 (0.163)	43.82 (45.19)	0.997 (0.033)	5.08 (10.56)	0.987 (0.065)	7.59 (15.33)	0.897 (0.154)	19.96 (23.63)
Chi-square	$t = 0$	0.970 (0.095)	13.72 (17.91)	0.993 (0.047)	8.98 (9.25)	0.927 (0.146)	26.56 (34.23)	0.947 (0.122)	19.01 (27.31)	0.867 (0.194)	27.38 (34.06)
	$t = 1$	0.960 (0.118)	18.90 (20.58)	0.823 (0.179)	51.85 (41.81)	0.913 (0.161)	28.66 (30.54)	0.927 (0.153)	22.02 (21.78)	0.710 (0.261)	43.65 (36.31)
Mixture	$t = 0$	0.933 (0.163)	20.74 (26.34)	0.973 (0.090)	15.82 (24.41)	0.843 (0.228)	43.57 (45.54)	0.857 (0.212)	38.07 (39.80)	0.837 (0.208)	27.11 (29.63)
	$t = 1$	0.900 (0.167)	31.38 (32.59)	0.787 (0.197)	55.48 (42.62)	0.723 (0.275)	62.13 (48.56)	0.813 (0.223)	46.34 (39.56)	0.747 (0.254)	41.56 (38.53)
Case II: $(n, p) = (200, 2000)$											
Normal	$t = 0$	1.000 (0.000)	3.00 (0.00)	1.000 (0.000)	3.01 (0.10)	1.000 (0.000)	3.18 (0.55)	1.000 (0.000)	3.00 (0.00)	1.000 (0.000)	3.30 (1.11)
	$t = 1$	1.000 (0.000)	3.81 (3.22)	0.713 (0.116)	368.10 (421.14)	1.000 (0.000)	3.21 (1.09)	1.000 (0.000)	3.92 (3.61)	0.973 (0.090)	15.40 (31.98)
Chi-square	$t = 0$	0.997 (0.033)	4.07 (5.62)	1.000 (0.000)	3.46 (1.06)	0.967 (0.111)	16.22 (34.53)	0.990 (0.057)	6.40 (17.25)	0.967 (0.111)	15.89 (28.73)
	$t = 1$	0.963 (0.104)	15.65 (25.17)	0.717 (0.128)	367.71 (375.10)	0.917 (0.152)	26.47 (32.79)	0.963 (0.104)	26.53 (62.36)	0.773 (0.230)	110.88 (200.32)
Mixture	$t = 0$	0.967 (0.100)	17.41 (41.13)	0.983 (0.087)	12.87 (35.70)	0.743 (0.270)	135.39 (235.60)	0.873 (0.187)	67.17 (157.79)	0.937 (0.154)	25.41 (48.49)
	$t = 1$	0.863 (0.195)	91.90 (190.58)	0.663 (0.160)	457.32 (465.87)	0.663 (0.318)	189.76 (275.11)	0.760 (0.275)	169.95 (289.15)	0.750 (0.255)	114.96 (184.37)

Note: The mean (standard error in the parentheses) values are based on 100 replications.

gRRCS is significantly affected by the correlation between the variables. Overall, LqG outperforms the alternatives in all the cases that were considered.

4 | REAL DATA ANALYSIS

4.1 | Lung adenocarcinoma data

Gene expression research plays an important role in biomedical science. Genes that share common biological functions, chromosomal locations, or regulations can be viewed as a set; this is called a gene set. GSEA is a powerful analytical method as shown in Subramanian et al⁴ and Efron and Tibshirani.⁵ This section presents the analysis of lung adenocarcinoma (LUAD) data in the Cancer Genome Atlas (TCGA) database, comprising 229 samples and 18 316 genes. LUAD is a collective subtype of non-small-cell lung cancer (NSCLC), which is the most common type of lung cancer. First, we

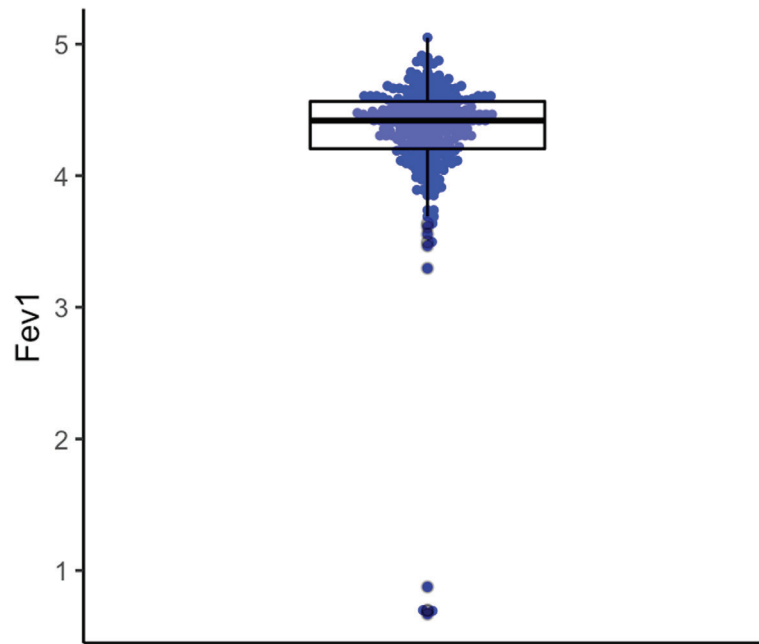


FIGURE 2 LUAD data analysis: Beeswarm plot of logarithm of Fev1, a point indicates an observation and the boxplot is displayed [Colour figure can be viewed at wileyonlinelibrary.com]

exclude the genes that have a single unique value, and the remaining 6945 genes can be divided into 609 gene sets according to the potential targets of regulation by transcription factors. Additional details can be found in MSigDB (<http://software.broadinstitute.org/gsea/msigdb/index.jsp>). Note that the gene sets in MSigDB have overlaps, and an appropriate adjustment is made here in accordance with the nonoverlapping assumption. Specifically, when a gene belongs to more than one set, it is assigned to the smallest set, which ensures that each gene is assigned to only one group. There are one to 70 genes per set with a median of 11, which indicates that the number of covariates varies among the groups. The correlation within the groups is slightly larger than that between the groups.

4.1.1 | Fev1 as the response

The forced expiratory volume in one second (Fev1) is a major clinical measure of the lung function, which is a continuous variable for the response. It is analyzed on the logarithmic scale. The beeswarm plot of Y in Figure 2 indicates that the distribution of the response has a heavy tail and there exist some potential outliers in the response. This served as motivation to apply a robust method to conduct group screening for this dataset. To identify the important gene sets of Fev1, group screening was performed and $\lceil n/\log(n) \rceil = 42$ gene groups were screened out. The detailed results of each method are listed in the Section B of the supplementary material. The results of LqG, Lq1, and gRRCS have the most overlaps.

The following describes the gene sets identified by the different methods from the perspective of biomedical research. The first is IQGAP2, which is screened out by all of the methods. IQGAP2 is a member of the IQGAPs (IQ Motif Containing GTPase Activating Proteins), which are evolutionarily conserved proteins. Kumar et al.²⁰ analyzed the mRNA expression of IQGAP2 in individual cancers in the TCGA database, and determined that the IQGAP2 transcript levels were reduced across different cancers, which indicated the potential of tumor suppression. The second is a gene set that contains the genes EHBP1 and GPR161, which are identified at the top rank by LqG, Lq1, and gRRCS. Both of these genes have an association with cancer. In particular, GPR161 is a drug target for breast cancer as suggested by Feigin et al.²¹ The third is OTUD7B, which is also identified by LqG, Lq1, and gRRCS at the top rank. OTUD7B is a multifunctional deubiquitinylase, which plays an essential role in inflammation and proliferation signals. It has been recently found to be highly expressed in lung squamous carcinoma and adenocarcinoma, and it correlates with a worse prognosis.²² The last is Cyclin-dependent kinase 1 (CDK1), which is only identified by LqG. It is an adverse prognostic biomarker for LUAD, and its increased expression could indicate a high risk of cancer recurrence and poor survival chances for patients with LUAD.²³

TABLE 5 Ten-fold cross-validation error in the LUAD data based on 100 random splittings

Method	RMSE	Standard error	Group size	Model size
LqG-SCAD	0.498	0.030	2	10
Lq1-SCAD	0.517	0.020	3	16
gRRCS-SCAD	0.524	0.020	4	19
gHOLP-SCAD	0.580	0.022	4	31
gSIS-SCAD	0.577	0.024	5	36

TABLE 6 Results of the LUAD data simulation

	LqG		Lq1		gRRCS		gHOLP		gSIS	
	PC	MS	PC	MS	PC	MS	PC	MS	PC	MS
Normal	1.000 (0.000)	32.06 (1.43)	0.917 (0.144)	41.24 (1.91)	0.713 (0.116)	44.98 (2.58)	1.000 (0.000)	5.09 (0.29)	1.000 (0.000)	27.29 (1.72)
<i>t</i>	1.000 (0.000)	32.51 (3.12)	0.857 (0.165)	42.32 (4.28)	0.720 (0.122)	48.79 (6.29)	0.990 (0.074)	7.88 (11.90)	1.000 (0.000)	28.31 (3.73)
Cauchy	0.967 (0.100)	32.59 (7.40)	0.817 (0.166)	44.69 (9.73)	0.730 (0.131)	52.19 (11.40)	0.560 (0.386)	171.36 (173.64)	0.677 (0.335)	122.37 (138.30)

For a comparison of prediction performance, the methods are examined via 10-fold cross-validation. The screened out groups are then selected by group SCAD and refitted via OLS for prediction purposes. We report the means and standard errors of the root mean square errors (RMSEs) of the prediction based on 100 random splittings and the median of the final chosen model size in Table 5. It is observed that the model selected by the LqG-SCAD is the smallest and it achieves the smallest prediction error.

4.1.2 | Real data simulation

As it is difficult to assess the accuracy of the screening with actual data, this study applies a real data simulation. This is achieved by generating the response values when using the genetic data from 229 LUAD patients. This study evaluated the screening accuracy of the proposed method under the group structure of this dataset. Each gene was standardized to have a mean of 0 and a variance of 1. Three gene sets were randomly selected as true significant groups with group sizes of seven, four, and nine, respectively. The true coefficients are set as S1. Then, the response Y is generated following a normal distribution, a $t(2)$ distribution, and a Cauchy distribution, respectively. We screened out the same size of the $\lceil n/\log(n) \rceil = 42$ groups, and examined their performance based on 100 replications. The results are summarized in Table 6, and indicate that the LqG can always screen out the important gene sets. Note that gHOLP and gSIS work well in the normal distribution and t distribution; however, they exhibit a worse performance when the response has a Cauchy distribution.

4.2 | Mammalian eye diseases

We used the dataset reported in Scheetz et al²⁴ to illustrate the application of the proposed screening method in a non-parametric setting. This dataset was also analyzed for nonparametric screening in Fan et al.¹⁷ The dataset comprises data of 120 twelve-week-old male rats selected for tissue harvesting from the eyes and for microarray analysis. The microarrays used to analyze the RNA from the eyes of these animals contained 31 042 different probe sets (Affymetric GeneChip Rat Genome 230 2.0 Array). Following Scheetz et al, we selected 18 976 probes as they exhibited a sufficient signal for

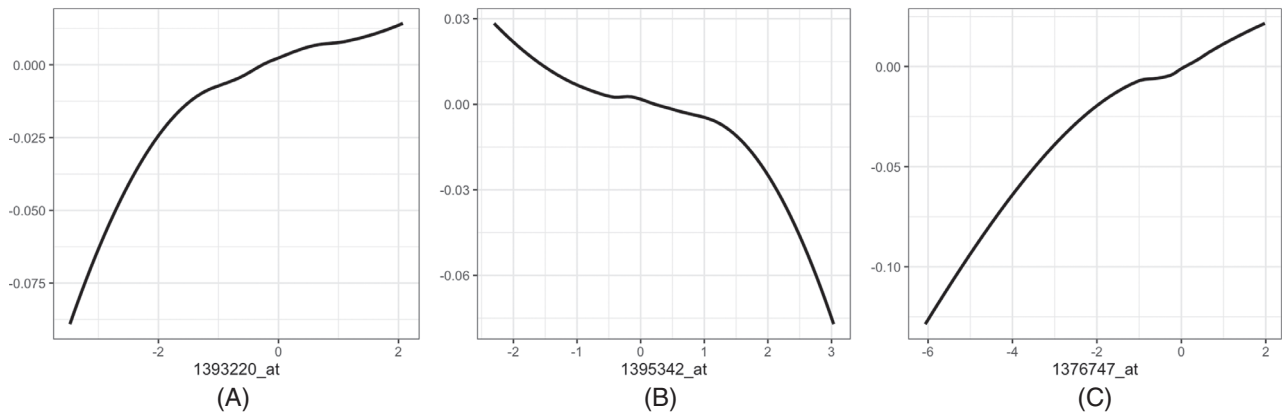


FIGURE 3 Mammalian eye diseases: Loess functions for the probes

TABLE 7 Ten-fold cross-validation error for the eye disease data based on 100 random splittings

Method	RMSE	Standard error	Model size
LqG	0.013	0.008	3
gRRCS	0.038	0.021	4
gHOLP	0.058	0.108	15
gSIS	0.024	0.014	4

a reliable analysis and at least a 2-fold variation in the expressions. We focused on the top 5000 probes with the highest sample variance. The gene expression levels were analyzed on a logarithmic scale.

4.2.1 | TRIM32 as the response

Similar to Huang et al,²⁵ we are interested in finding the genes that are related to the TRIM32 gene, which causes the Bardet-Biedl syndrome,²⁶ a genetically heterogeneous disease of multiple organ systems including the retina. The loess fitted functions of the predictors and response are shown in Figure 3, and the curves have a nonlinear trend. To identify the important probes that may have a nonparametric effect on TRIM32, we conducted group screening. The basis functions of each component were viewed as a group and we screened out $\lceil n/\log(n) \rceil = 25$ groups of basis functions, that is, 25 probes. The detailed results are listed in the Section B of the supplementary material. The probes screened out by various methods differed. One of them, probe 1376747_at, which was also selected in Fan et al, was screened out by all of the methods.

To assess the prediction performance of the methods, we further conducted variable selection for the screened out 25 probes with GAMSEL (generalized additive model selection).²⁷ The prediction of each method was assessed via 10-fold cross-validation. The means and standard errors of the RMSE of the prediction based on 100 random splittings and the median of the final chosen model size are reported in Table 7. LqG achieves the smallest prediction error.

4.2.2 | Real data simulation

Three genes were randomly selected as true significant nonparametric effects, and they are referred to as X_1 , X_2 , and X_3 without loss of generality. We generated the response Y from the following model:

$$Y = 3g_1(X_1) + 9g_2(X_2) + 5g_3(X_3) + \epsilon,$$

TABLE 8 Results of simulations with the eye disease data

	LqG		gRRCS		gHOLP		gSIS	
	PC	MS	PC	MS	PC	MS	PC	MS
Normal	0.993 (0.047)	9.98 (28.65)	0.663 (0.145)	332.10 (230.61)	0.973 (0.090)	27.30 (31.53)	0.993 (0.047)	8.97 (30.20)
Chi-square	0.853 (0.223)	98.55 (164.05)	0.620 (0.226)	360.51 (294.77)	0.813 (0.165)	169.23 (228.55)	0.837 (0.233)	121.64 (187.58)
Mixture	0.780 (0.303)	124.02 (172.37)	0.570 (0.268)	437.31 (338.18)	0.753 (0.167)	260.59 (355.10)	0.677 (0.338)	206.87 (286.05)

where $g_1(x) = (2x - 1)^2$, $g_2(x) = \sin(\frac{1}{2}\pi x)$, and $g_3(x) = x$. We considered different distributions for ϵ , including the standard normal, $\chi^2(2)$, and a Gaussian mixture distribution with the density function $f_\epsilon(x) = 0.8\phi(x; 0, 1) + 0.2\phi(x; 0, 25)$. Since each nonparametric component can be expanded as a group of basis functions with $K = \lfloor n^{1/5} \rfloor = 3$, we conducted group screening on the basis functions. We screened out the same size of $\lfloor n/\log(n) \rfloor = 25$ genes, and examined their performance based on 100 replications in Table 8. LqG has a better performance than the others, especially gRRCS, which do not apply nonparametric screening directly, particularly in the case of correlated predictors.

5 | DISCUSSION

For the screening analysis of ultra-high-dimensional data, a robust approach is in great demand due to the common occurrence of data contamination. Moreover, when variables are grouped, ignoring the group structure and directly applying individual variable screening may not be recommended. In this study, the Lq group (LqG) screening method was developed to accommodate the grouped variables and data contamination. This method conducts group screening by ranking the marginal utility of each group, which is defined based on the maximum Lq-likelihood estimates of the regression using each group of variables. By inheriting the advantage of the MLqE in small or moderate sample situations, the LqG is more robust to heterogeneous data and heavy-tailed distributions. The simulation demonstrated that the proposed approach has a competitive performance, even when there is a high correlation within or between the groups. In addition, the proposed method can be applied to screening for the nonparametric component as a special case.

The proposed method focuses on a nonoverlapping group structure. In practice, modeling complex group structures are common in scientific fields, especially in medical and genetic studies. Deriving methods that enable the identification of important groups with complex group structures for ultra-high-dimensional data is an important task for future research. Furthermore, the cutoff value is determined via hard thresholding in the proposed method; it will be of interest to study the data-driven rule for the cutoff value¹³ in future studies.

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DATA AVAILABILITY STATEMENT

The datasets we used in this paper are publicly available on <http://www.cbioportal.org/> and www.ncbi.nlm.nih.gov/geo.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

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